CHAPTER 7

Infrastructure

1. PUBLIC PRIVATE PARTNERSHIP

There has been growing emphasis on private sector participation in all sectors of infrastructure and the provision of infrastructure has been the priority sector of the Government. The Government of Goa is actively pursuing the Public Private Partnership (PPP) model to meet the gaps in the provision of infrastructure services.

The PPP Cell headed by a Director and directly monitored by the Chief Secretary is functioning since July 2006 in the Secretariat to monitor progress of various projects initiated by the Government under PPP mode. The first phase of the first project under PPP concept "The Goa Broad Band Network" Project (GBBN), a unique ultra high speed network was commissioned at the hands of Hon'ble Prime Minister on 27th December 2007. Setting up of Government Portal and State Data Centre which will be linked to GBBN project is also in progress.

The Cabinet has also accorded in principle approval to a large number of projects under PPP mode in the various sectors such as Transport, Urban Development, Health, Tourism and Industries and further action on project development has been initiated. Some of the important projects identified are construction of modern Bus Stands, Truck Terminals, setting up of an Oceanarium, Solid Disposal Facility, Superspeciality Block at Goa Medical College and Tourism related projects.

Inorder to formulate an appropriate Infrastructure Policy, Sector specific strategies and identification of major infrastructure projects under PPP concept, appointment of a Professional Advisory Group for the State is under the consideration of the Government.

2. POWER

The State of Goa does not have generation of its own except for one 48 MW Naptha based power plant of M/S Reliance Salgaonkar Power Company Ltd. which went into commercial operation in August 1999. Thus, Goa is mostly dependent on the power allocated from the Central Generating Stations in the Western and Southern Regions wheeled through the neighboring

State Grids. The present allocated share from these Central Generating Stations is 293 MW from the Western Region and 100 MW from the Southern Region. In addition, an additional 10 MW has also been allocated from the unallocated share only during the peak hours.

The power from both the Western and Southern Regions is received at the 220/110/33 KV Sub-Station at Ponda in Central Goa, at 220/110 KV Sub-Station at Tivim in North Goa and at 220/110 KV Sub-Station at Xeldem in South Goa. Within the State, it is transmitted from Ponda Sub-Station to Verna and Kadamba 110/33 KV Sub-Stations and further distribution is done through Sub-transmission and Distribution network consisting of 45 numbers of 33/11 KV Sub-Stations and 3,800 distribution transformers.

The present maximum demand of Goa is of the order of 400 MW including drawal of power from Reliance Energy Limited (REL), IPP. Goa, is however, forced to overdraw some power from the Western Region due to high peak load. Goa also buys power from the co-generation power plants of M/s Goa Energy Pvt. Ltd. (GEPL) and M/s Goa Sponge Pvt. Ltd. (GSPL) to the extent of around 14-21 MW and 1 MW respectively.

The Government took a decision to award works worth Rs. 60.00 crore on deposit basis to IRCON, a Government of India Undertaking, functioning under the Ministry of Railways. The scope of the work includes erection of 220/33 KV, 3x50 MVA Sub-Station at Cuncolim alongwith the 220 KV D/C line from Xeldem to Cuncolim to feed this proposed Sub-Station.

The feasibilty studies for erection of 220/110/33/11 KV Sub-Station at Verna and also interlinking Ponda, Verna, and Xeldem Sub-Stations with a 220 KV line are also being carried out and the same is awarded to POWERGRID.

a. Various works/measures undertaken

- An additional 40 MVA power transformer alongwith other swithchgear equipment has been commissioned at 110/33 KV Sub-Station at Verna. This has enabled to release additional loads in the Industrial Estates.
- The work of providing 40 MVA power transformer at 110/33 KV Sub-Station at Ponda alongwith other swithchgear equipment has also been completed, which will augment the transformation capacity at Ponda Sub-Station by another 40 MVA.
- A 33/11 KV, 6.3 MVA Sub-Station at Sanguem has been commissioned adding to the transformation capacity and reducing the long feeder lengths in the area and in turn improving the voltage profile. A similar Sub-Station at Poinguinim has also been completed and will be commissioned on completion of the 33 KV line feeding the same, which is under erection.
- 33/11 KV, 6.3 MVA Sub-Stations at

Cavelossim, Shiroda and Campal is also nearing completion and the Sub-Station at Ponda is under execution.

- The land acquisition process for erection of 220/33 KV, 3x50 MVA Sub-Station at Cuncolim is in the final stage. The preliminary works in respect of the Sub-Station and line is in progress.
- The NIT for providing additional 2x40 MVA, 110/33 KV transformers at Kadamba 110/33 KV Sub-Station alongwith new 110 KV line is under preparation and the work will be tendered shortly.
- The underground conversion works in Margao and Panaji have almost been completed except for a few pockets where work is in progress.
- The notification for giving further incentives to bulk consumers like HT consumers and EHT consumers for achieving better power factor and higher load factor subject to provision of TOD metering and keeping of harmonics within permissible limits, has been published. Some of the major consumers from these categories have taken action for providing TOD metering and also keeping harmonics within permissible limits. However, the incentives will be released only upon verification/satisfaction of the conditionalities.
- During the year, around 43 MW of Captive/Co-generation has been streamlined to help overcome peak hour power shortfalls and earn additional revenue through sales. Purchase of peak hour power was also arranged and by these measures, load sanctions to the tune of 30 MW could be made.

b. Steps taken towards mobilization of resources

- Taking advantage of the Availability Based Tariff (ABT) regime, the Department by proper scheduling of declared availability and load management has been earning considerable revenue by way of pooled unscheduled interchange charges (UI charges) levied on other defaulting States.
- By strict implementation of the financial reduction plan through the third Supplementary Agreement executed with RSPCL, the Government has considerably reduced the financial burden on drawal of power from RSPCL. The present drawal from RSPCL has been brought down from 39.8 MW to 15 MW.
- Monthly revenue collection targets were fixed for all the O&M Divisions and watched carefully. Billing, revenue collection and disconnection of installations are observed strictly. Surprise inspection of installation by MRT Vigilance wings has been speeded up and it is yielding good revenue, by curtailing the theft of energy. Theft cases are detected every month and the same fetches additional revenue. The revenue is being watched on day-to-day basis for effective control. Even recovery of dues from other Government Departments

has been resorted to

- The electricity duty has been increased from the existing 2 paise/unit to 5 paise/unit, resulting in additional resource mobilization.

c. New initiatives to improve the service to the people

- For improving the service to the people, the Department is benchmarking certain parameters and fixing targets for improvement of these parameters and implementation of APDRP. The schemes and measures are under implementation and have improved the quality and reliability of power to the consumers.
- The mobile vans have also been outsourced alongwith the drivers for round the clock service, for the first time. Thus, it has been possible to increase the number of mobile vans in service in the urban areas and also has helped in providing mobile van service to the rural areas.
- The department has also provided Operation & Maintenance Division with trolley mounted mobile transformers to enable the power supply to be restored within a couple of hours to the locality in case of breakdown of the distribution transformer centres even at night. This greatly improves reliability of power supply.
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d. Major achievements during 200-08 (upto Nov.'07)

- The new Sub-Stations at Sanguem and Poinguinim taken up under DPR Phase III have been completed. The other underground conversion works in the coastal areas of North Goa and some feeders in South Goa taken up in the same phase are under execution alongwith the balance works of DPR Phase- III.
- The Sub-transmission and distribution improvement works taken up in South Goa under Phase-I is also nearing completion.
- The underground conversion works in Margao and Panaji have almost been completed except for a few pockets where work is in progress.
- The work of providing 2nd 40 MVA power transformer at Verna 110/33 KV Sub-Station has also been completed and commissioned during the year.
- The work of providing an additional 40 MVA power transformer at Ponda 110/33 KV Sub-Station has been completed.
- The work of erection of 220/33 KV, 2x50 MVA Sub-Station at Amona is progressing and the 33 KV portion will be commissioned shortly.
- The land acquisition process for the 220/33 KV, 3x50 MVA Sub-Station is in progress. The preliminary works associated with the Sub-Station has been taken up.
- The pre-qualification of tenderers for the work of providing 2nd 100 MVA power transformer at 220/110 KV Sub-Station at Xeldem is under finalisation.
- Additional allocations of 51 MW from Central Generating Stations has also

been arranged.

- It is proposed for the first time to have a power banking arrangement for exchange of upto 50 MW off peak power (00.00 to 06.00 hours) for evening peak hour power (18.00 to 22.00 hours).

e. Measures for energy audit and reduction of losses

Under the APDRP scheme, the Distribution Transformer Centres (DTC) have been/are being provided with meters. Already all the feeders at Sub-Stations upto 11 KV have been provided with electronic meters. With the DTC metering when fully completed, energy audit can be carried out feederwise and losses will be plugged. At present, feeder losses right from 220 KV level upto 11 KV level are being monitored on monthly basis.

Replacement of non-working meters for LT consumers by State-of-the-art electronic meters has been taken up in full swing to ensure accurate metering and bring down the losses.

f. Human Resource Development

- The Training Centre at Ponda has now started imparting training to the newly recruited line staff of the Department, so that new recruits are trained before entering service.
- A special Drum Training Programme with outside faculty has been organized to act as a refresher course for senior line staff who can then be used as trainers for junior line staff.
- 30 engineers have been enrolled for a 6-month part time Advanced Certificate Course in Power Distribution Management course through IGNOU. This batch has also answered the exams conducted by IGNOU in December '07.

3. TRANSPORT

a. Road Transport

As on December 2007, the number of Motor Vehicles registered in the State stands at 6,11,615. On an average over 45000 vehicles are being registered every year. The number of vehicles registered during 2007-08 (upto Dec, 2007) is 32844. It is evident that about 71% of vehicles are in the category of two wheelers followed by cars and jeeps including Taxi which account to about 20%. The number of motor vehicles registered during the period from 1996-97 to 2007-08 is given in Annexure 25. The number of driving licenses issued as on 31st December 2007 according to place of registration is given in Annexure 26.

Achievements

i) Smart Card driving license and Dealers level registration of Vehicles

The department provides easy and efficient services to the public viz. registration of new vehicles through authorized dealers, Smart Card driving licence which is durable, easy and elegant to carry. These services have been commissioned at the RTO Panaji, Margao, Mapusa, Vasco, Bicholim, Quepem

and Ponda. Under this project, 36170 people have been issued the Smart card driving licenses upto 30/11/07.

ii) Subsidy for the purchase of Yellow/Black Motor Cycles/Auto-rickshaws/Taxis

The Government has continued the implementation of the scheme for grant of subsidy for replacement of buses and minibuses older than 15 years, in order to curtail vehicular pollution and to promote clean environment. Under this scheme, during the year 2007-08 (upto Nov. 2007) Rs.35.25 lakh has been disbursed. Similarly, the department has launched a new subsidy scheme for replacement and procurement of new yellow/black motorcycles, yellow/black auto rickshaws and yellow/black taxis to protect environment and to enhance road safety.

iii) Pollution Under Control (PUC)

The department has introduced upgraded schemes of Pollution Under Control (PUC) Certificate status. Under the scheme, all stations authorized to issue PUC certificates for vehicles have been upgraded with modern computerization equipment facilities to test the pollution levels of a vehicle. 26 PUC centres have been issued licence under this scheme.

iv) Moratorium on issue of tourist permits

To regulate the growth in the number of tourist vehicles and to promote a healthy competition in the transport sector, the Government had imposed a moratorium on issue of tourist permits. However, on account of the ensuing tourist season a need was felt to service the tourism industry and accordingly, the Government decided to open up grant of permits for buses/mini buses/taxis/maxi cabs having a carrying capacity of 1000 c.c. and above.

v) Road Safety Programmes

To minimize road accidents in the State, the department has launched vigorous publicity through the print and electronic media. Awareness programmes and counseling sessions are going on, with effect from the month of November 2007. The training programme for all the bus drivers and conductors has been commenced for a period of six months. During the year 2007-08 (upto November 2007), the department has imparted training/counseling to heavy vehicle drivers. The department has completed the making of short films on three road safety themes namely overloading, overcrowding and over speeding and these will be shown on all T.V. Channels and in Cinema Theatres throughout the year. Medical camps for eye checkup for drivers has been conducted during road safety week 2007 and awareness programmes on road safety are also being conducted at the level of school and colleges.

vi) Construction of Modern State-of-the-Art Bus terminals and Bus Stands in Goa

It is proposed to construct modern bus stands in Goa with adequate infrastructure facilities. It is proposed to construct modern State-of-the-Art Bus terminals at Margao and Mapusa. Sufficient land is already available at Margao and in the case of Mapusa, additional land of 20,000 sq.mts. is being acquired for the expansion of the Mapusa bus stand. Bus stands are also proposed at Shiroda, Valpoi, Honda and Pernem. Land of 6383 sq.mts. for Shiroda bus stand has been acquired in Ponda taluka. Land admeasuring an area of 1450 sq.mts. has also been acquired for Honda bus stand in Sattari taluka. For construction of bus stand in the village of Valpoi in Sattari taluka land measuring an area of 17500 sq.mts. is under acquisition. Section VI, Notification is yet to be issued under Urgency Clause. In case of Pernem Bus stand land acquisition process is going on. The work of construction of these Bus Stands is entrusted to Goa State Infrastructure Development Corporation (GSIDC).

vii) High security registration plates

The process of executing legal documents for implementing the scheme of "High Security Registration Plates" is in the final stage. The tender in this regard has been finalized.

viii) Speed Governors

With a view to minimize road accidents in the State and to control overspeeding, the Government has taken mandatory steps for owners of specific category of vehicles, especially the heavy commercial ones to fit Speed Governors.

ix) Mobilisation of additional resources

The Directorate of Transport has taken initiatives towards mobilization of additional resources and during the current financial year 2007-08 upto December 2007, revenue of Rs.66.56 crore has been collected. Also under Goa Rural Improvement and Welfare Cess Act, 2000 and the Rules 2006, an amount of Rs.8.51 crore has been collected during the current financial year 2007-08 upto November 2007.

x) Full Fledge RTO Office at Quepem

To provide easy and efficient services to the public of Quepem, Sanguem and Canacona talukas, the Regional Transport Authority has been constituted w.e.f. 23/08/2007 wherein the office has been upgraded to a full fledged one.

b. Inland Water Transport

Captain of Ports is the implementing Authority for Inland Vessel Act, 1917, the Goa Daman & Diu Barge Tax Act, 1973, the Goa, Daman & Diu Barge (Taxation on Goods) Act, 1985, the Goa, Daman & Diu Port Rules, 1983 and the Indian Ports Act, 1908.

The International Ship and Port Security Cell at Panaji (non major Port) has been declared ISPS Compliant w.e.f 31/03/2005. An annual fee of Rs.20,000/- is mandatory towards operation and maintenance of the communication centre set at the Director General of Shipping Office in Mumbai. Besides, an annual fee of Rs.25,000/- p.a. is also mandatory towards audit of Port operation. In addition to the above, expenditure of approximately Rs.45.00 lakh towards standard Shipping Ports Communication/Security equipment/ accessories/ consumable, Annual Port Radio Station Licence fees, periodical Port ISPS Security meetings, security surveillance of the port area, professional man power etc. has to be incurred since the port is declared as ISPS Compliant Port.

The Marine workshop maintains and repairs 37 ferries and one motor launch of the department which operate on19 ferry stations all over Goa. Minor repairs to the vessels are being carried out in the department for which material required is purchased and stocked in the stores of the department.

Initiatives to improve Service to the people

- The River Navigation Department provides/assures safe, dependable and affordable water transport facilities at different ferry routes in the Inland Waters of Goa. The department has maintained free passage via ferry for passengers w.e.f 1.4.2000 and for two wheelers w.e.f 1.1.2006.
- Inorder to provide better service to the public, Government has proposed for a three-shift system work pattern for the flotilla staff of the department.
- Tender has been floated for procurement of two new ferries at a cost of Rs.90.00 lakh approximately and proposal for another additional four is being placed for approval. Proposal for major repairs of three ferries through private agencies has also been approved and the process of awarding the work is in progress.

Achievements

- The department is leasing the toll fee collection on all the ferry station in Goa except Kiranpani Arronda which is an inter-State route.
- Government has approved the registration of all the vessels belonging to the River Navigation Department which were not registered till date.
- The works of the Upgradation of Panaji minor ports and connecting jetties at Old Goa and Britona have been entrusted to the Goa State Infrastructure Development Corporation. Preliminary soil testing of riverbed has been carried out by the consultant.
- The Hydrographic survey of the selected Inland Waterways of Goa through the Minor Ports Survey Organization has been completed for rivers at Mapusa, Chapora & Sal.
- The 45th, 46th and 47th batches of new entrants course in Deck Engine

discipline is completed on 8th September 2007, thus providing skilled trained work force to cater to the need of barge industry. Following the same pattern, the school has now plans to conduct 48th, 49th & 50th batch of 15 days and 5 days on & off basis from 7th January 2008.

- Two new wreck marker buoy procured for the development at the cost of Rs.3.56 lakh.
- Construction of Grab dredger is in progress.
- The department collects adequate revenue through various dues, log book, dispensation and endorsement of vessels, survey fees, licence fees of vessels, barge tax etc. and through hire of riverine land.
- A new ferry route at Manke-Tar (Shiroda) to Ullando (Cortalim) is approved for operation for which ramps are under construction.
- A new ferry ramp at St.Estevam is proposed for construction which will reduce the distance between Bicholim and Marcela.

c. Roads

Following major spill over works are completed for the year 2007-08, upto November 2007.

Sr. No	Name of work	Estimated cost (Rs. in lakh)
1	Improvement & Widening of road by providing hotmix carpet from Agarwada Junction to Pernem Town (MDR-19) in Pernem Constituency	294.84
2	Providing & hot mix carpet (MDR-16) along Malim-Betim road via Salvador-do-Mundo Church in Aldona Constituency	84.09
3	Improvement & B.T. of various roads in V.P. Mandrem under Mandrem Constituency	68.71
4	Construction & B.T. of road from Manshiwada to Balkhazan via Tumbwada in V.P. Corgao in Pernem Constituency	56.26
5	Providing hotmix carpet for various roads under V.P. Virnoda and V.P. Porascodem in Pernem Constituency	59.21
6	Providing hot mix carpet for the road from Bomwada to Arabo bridge and roads leading to Bomwado-Baghbandir and Don Bosco High School under VP Tuem in Pernem Constituency	55.33
7	Improvement of various roads in V.P. Ozarim by providing hot mix carpet (phase-I) in Dhargal constituency in Pernem Taluka	76.08
8	Construction of left out roads in Corjuvem and Calvim Islands and other Wards of V.P. Aldona constituency	60.71
9	Providing 40 mm hotmix carpet for internal roads of Pomburpa Olaulim in a length of 7.90 kms in Aldona Constituency	52.14

10	Improvement & providing 40mm thick bituminous concrete to the entire roads in journalist colony in V.P. Penha-de-Franca in Aldona Constituency	66.54
11	Construction of left out roads in Socorro in Aldona Constituency	52.19
12	Providing hotmix carpet on MDR 28 from Barazan to Kankirem upto Sattari	63.03
13	Widening of MDR 27 Pissurlem Wadakade to Advoi in Poriem Constituency	45.34
14	Providing 22mm thick mix seal surface to road from Ambedem to Codal village in V. P. Nagargao	78.29
15	Improvement of road from Mungal to Betalbatim church under Benaulim & Loutolim	66.97
16	Improvement of riding quality of road from Panzorconi to Bhatim in a length of 3.5 kms.	94.46
17	Strengthening & resurfing of main road of Touristic imp. In V.P. Colva	51.00
18	NH 17 junction at Dabolim Park Plaza in VP Cortalim const. under Touristic importance road	139.09
19	Periodical renewal of Existing carriage way in km 28/00 to km 33/00 NH-17 (Panaji-Panaval Section) in the State of Goa	136.67
20	Providing hot mix carpet to the road from Sateri Temple at Curti to Hotel Menino junction at Ponda i.e. from km. 122.50 to 125.00 km in Ponda Constituency	81.14
21	Improvement of road from Rajiv Kala Mandir to Bethora bye-pass in Ponda Constituency	61.42
22	Construction & BT and providing AC carpet and premix for various road at Mattawada, Gaodongor, Bhatwada, Mahalewada, Vithoba Temple, Satore-wada, crematorium and Kapdshem and Wagaped and Thal area in V.P. Shiroda of Shiroda Constituency	81.82
23	Providing hot mix carpet to the various roads at Tiral, Udiwada, Takawada and Palwada in V.P. Usgao Ganjem in Ponda Constituency	107.04
24	Providing hot mix carpet for internal roads in V.P. Mollem in a combined length of 11.10 kms. in Sanvordem Constituency	91.40
25	Improvement to various roads in V.P. Curti Khandepar of Ponda Constituency	143.59
26	Widening and hotmixing of road from Tulsimola to Keri in V.P. Keri	89.03
27	Widening of MDR 27 Pissurlem wadakade to Advoi	50.61
28	Widening and imp. of the riding surface on SH4 between ch 12.00 to 26.50 km. on Keri Belgaum road	246.62
29	Improvement and asphalting of various roads in Pale	105.37
30	Improvement and hotmixing of Phalwada road in V.P. Cudnem in Pale Constituency	49.98
31	Providing hot mix to the road Assonora Bicholim SH upto Muslimwada in a length of 7.00 k.m.	64.11
32	Improvement of Assonora Advalpal road in a length of 6.66 kms.	74.69
33	Providing hotmix surfacing of the road from Chamarkhond to Maulinguem and upto Talewada Bridge in Mayem Constituency	68.41

d. National Highways

The upgradation of NH-17 to 4-Lane is being undertaken by MOSRT & H-New Delhi (Ministry of Shipping Road Transport and Highways) under NHDP-phase III programme. This portion also includes construction of parallel Zuari bridge, Talpona and Galgibag bridges, portion of highway NH 4A in Goa from Anmod to Panaji for 4 lane is also being identified by MOSRT & H under the above programme.

4. WATER RESOURCES

The Water Resources of the State has been assessed at 8570 M.m3 by the Central Water Commission (CWC). The storage and water distribution for irrigation is made through various major, medium and minor irrigation projects. Considering the need of water by 2050, a Master Plan for Mandovi, Zuari, Talpona and Galgibag river basins is prepared by a panel of Experts.

I. Major and Medium Irrigation

a) Salaulim Irrigation Project

The project has been completed except for matters related to rehabilitation and land acquisition. A total potential of 14106 Ha. has been created. The Project has a Water supply component of 160 MLD for domestic and Industrial use. In addition, permission has been granted to the PWD for use of 220 MLD of water for the expansion of Salaulim water supply scheme. 28 Water Users' Association (WUAs) have been formed, out of which 15 WUAs have been handed over to the charge of water distribution networks.

b) Tillari Irrigation Project

This inter-State joint venture of the Governments of Maharashtra and Goa envisages creation of culturable command area of 14,521 Ha. During the current financial year 2007-08, an amount of Rs. 30.00 crore has been allotted to the Goa Tillari Irrigation Development Corporation (GTIDC) for speedy completion of the project. The project is expected to be completed by May 2009 and supply 115.03 Mm3 of water to the water supply schemes at Assonora and Chandel. Upto March 2007, 1,246 Ha (Ultimate) potential has been created under the project. CAD works under the project have been taken up by motivating the farmers and Water Users' Associations are formed in the command of B/1 and B/2 distributaries of RBMC and Sal and Vadaval distributaries of LBMC. So far, 13 societies have been registered under the Goa Co-operative Societies' Act. The balance canal works are in progress. The present work status of the project as on 31-03-2007 is as follows:

Table 7.1: Status of the work of Tillari Irrigation Project

Sl. No.	Component of the Project	% of Completion	
	In Maharashtra		
I)	**Head Works**	100 %	
1	Main Dam (Earthern)	99%	
2	Masonary Dam	41.16%	
3	Tail Channel	100%	
4	Irrigation cum Power Outlet	100%	
5	Terwanmedhe Pick up Weir	100%	
6	Left Bank Main Canal	100%	
7	Right Bank Main Canal	100%	
8	Banda Branch Canal	50%	
	In Goa		
		Earthwork	**Lining**
II)	**Name of Canal**		
1	Right Bank Main Canal	93.82%	88.98%
2	Left Bank Main Canal	73.84%	57.47%
3	Sanquelim Branch Canal	100%	100%
4	Distributaries on LBMC	18.53%	16.82%
5	Distributaries on RBMC	29.77%	28.65%

Under the Accelerated Irrigation Benefit Programme (AIBP), grants to the tune of Rs. 1.91 crore have been received from the Government of India for the Tillari Irrigation Project during the year 2006-07. A proposal has been submitted to the Government of India to avail grants to the tune of Rs. 32.50 crore during 2007-08.

c) Anjunem Irrigation Project

This project has been completed in 1989 with a potential of 4,600 Ha. (Ultimate). Since the project is situated in high rain fall area of the Western Ghats which is prone to fast vegetation and weed growth, roots of these vegetation enter the canal lining and damage the concrete lining. Repairs and renovation of the existing canals in a length of 15.05 kms were planned in two phases during the years 2006-07 and 2007-08. Accordingly, 14 kms. length of canal has been renovated in the first phase. The second phase of renovation works of the lining of the main canals at a cost of Rs. 3.50 crore has commenced and the work is expected to be completed shortly. The Dam Safety Panel has suggested automization of the gates and replacement of the apron. The work of replacement of the apron is in progress and automization of the gate is being planned.

II. Water Development

- The Ministry of Water Resources, Government of India, has been implementing Hydrology Project Phase II by availing World Bank Assistance. Under this programme, the State of Goa has prepared a project of Rs. 22.64

crore. The work of construction of Laboratory level II plus and Data Centre has commenced. Training is being imparted to the Officials of the department in Water Quality Management, WISDOM software for Data Storage Centre and Hydro-meteorology etc. for implementation of this project.

- The Experts' Committee for Surface Water has so far approved 6 river gauge stations and another 4 are expected to be approved in its next meeting. A few ground water level observation stations have been identified and the same are being processed for approval of the Experts' Committee.
- Detailed survey and investigation of 17 irrigation projects in the State is in progress. Out of these, work on one project has commenced, 2 projects are in the final stage of design and remaining are in various stages of investigation.
- Minor Irrigation Projects have been initiated at Karvan and Gavanem in Canacona taluka. Design of the main dam and the head regulator is in advanced stage of finalization. The project will store water to the extent of 38.64 lakh cu.m (26.84 lakh cu.m. at Karvan and 11.80 lakh cu.m. at Gavanem) and will irrigate 400 Ha. (200 Ha each at Karvan and Gavanem) of command. Besides these two tanks, minor irrigation works such as construction of bandharas, improvement and renovation of springs, tanks, Lift Irrigation schemes and flood-protection works are being implemented. During the year 2007-08, irrigation potential of 245 Ha. has been created.
- Under the Western Ghats Development Programme, 47.00 Ha. have been covered by schemes such as open wells/tube wells/bore wells, Lift Irrigation schemes (L.I.S), tanks and bandharas.
- Boosted by the resounding success of the inter linking of rivers and post monsoon water harvesting programme, the State has undertaken construction of a series of bandharas for augmentation of water sources to water treatment plants at Opa in Ponda taluka, Assonora in Bardez taluka and Dabos in Sattari taluka, to help in ground water re-charge, for diversion of water and create storage in the riverine system as well as in the river banks by retarding the interflow of ground water into the riverine system. The construction of 93 bandharas has been completed on various rivers/nallahs so far. At present, construction of 14 bandharas holding a total storage of 14.00 lakh cu.m. is in progress. During the current year 2007-08, an additional 26 bandharas holding a total storage of 14.66 lakh cu.m. are planned for implementation.
- The work of inter-linking of Madei river at Ganjem to Khandepar river at Opa has commenced. The scheme will augment 30 MLD of water at Opa Water Works especially during the lean season of the year.

The present status of bandharas is as under:

Sl. No	Particulars	Number	Storage in MCM	Cost (Rs. in lakh)
1	Bandharas planned	133	42.12	9157.93
2	Bandharas completed	93	39.24	5236.15
3	Bandharas in progress	14	1.30	2784.46
4	Bandharas to be taken up	26	1.58	1137.32

The status of inter-linking of rivers is as under:

Sl. No	Name	Basin	Storage in M.Cum	Purpose		Remarks
				Irriga-tion	Water Supply	
1.	Inter Linking of Rivers	a) Chapora to Madei	3.125	-	3.125	Completed
		b) Zuari to Madei	12.50	-	12.50	Completed
		c) Ganjem to Opa	4.35	-	4.35	Work commenced

III. Command Area Development and Water Management (CADWM) scheme

- A proposal for availing central assistance from the Ministry of Water Resources, Government of India under CADWM programme has been submitted for approval. The estimated cost for the CADA works is Rs. 102.00 crore. Central assistance of Rs. 18.00 crore over a span of 5 years is expected from the Government of India.
- In response to the National Policy to encourage Participatory Irrigation Management (PIM), the State has prepared the PIM Manual and initiated action towards formation of Water Users' Associations and handing over operation and maintenance of network of distributaries and water courses coming under the command areas of Salaulim, Anjunem and Tillari Irrigation Projects. The State has made significant progress in this area by forming 57 Water Users' Associations registering the same under the Co-operative Societies' Act.

IV. Anti Sea Erosion Works

The Asian Development Bank (ADB) Management has approved a Project Preparatory Technical Assistance (PPTA) for preparing the "Sustainable Coastal Protection and Management Project" in the States of Goa, Karnataka and Maharashtra. The Ministry of Water Resources, Government of India has decided to implement the proposed project in the said States as a State Sector Project and has sought concurrence of the State Governments. The proposal is being examined. In the meantime, NABARD has also volunteered to provide financial

assistance for such schemes in the State of Goa and this proposal too, is being examined.

V. Urban Water Supply

A Master Plan is formulated to cater to the ever-increasing industrial, commercial, touristic and domestic demand for water supply and thereby to provide drinking water supply facilities on long term and sustainable basis. The augmentation water supply works at Opa, Sanquelim, Chandel are completed and commissioned. The augmentation works of Canacona WSS and Dabose will be completed shortly.

Various programmes to improve Water Supply in SC habitated locations and free water connections to SC in Betqui village are planned.

a. Scheme under external assistance for Augmentation of Water Supply & Sanitation

Inorder to cover the entire State with water supply, sewerage and sanitation for Mormugao, Margao, Ponda and coastal belts of North and South Goa, a scheme costing Rs.1031.90 crore out of which JBIC funding is Rs. 847.50 crore and State portion of Rs. 184.40 crore has been approved. The execution of the projects shall be completed by 2013. The loan agreement between Government of Japan and Government of India has already been signed on the 14th of September 2007.

The tender documents for selection of Project Management Consultants are in process. On completion of water supply project the dream of 24x7 safe sustainable water supply to four talukas namely Salcete, Sanguem, Quepem and Mormugao of South Goa will be achieved. Similarly, on completion of sewerage projects the hygiene situation of Margao, Mapusa and North coastal belt will be improved.

b. Improvement in existing water supply system in Salcete & Mormugao Talukas

The Water supply demand in the Port town of Vasco da Gama as well as industries in and around Sancoale is on the rise and contributes a major share to State revenue. A scheme comprising of 10000 cubic metres capacity MBR with suitable sump & pumps etc. is proposed at Vasco. Besides, a 10000 cu.m. capacity MBR is proposed to be constructed at Verna to take care of bulk consumers especially in the high bracket of revenue.

c. Augmentation of Rural and Urban Water supply from Salaulim impoundage

The present capacity of Salaulim Water Supply scheme is 160 MLD. In order to meet the increasing demand of water, it is proposed to augment the capacity of treatment plant at Salaulim by 200 MLD. The work is proposed in two phases of 100 MLD each. The work is also proposed to provide a parallel conveying main from Treatment

Plant to various locations. A concept proposal to augment water supply at an estimated cost of Rs.600.00 crore is under consideration.

5. URBAN DEVELOPMENT

A. MUNICIPAL ADMINISTRATION

a. Jawaharlal Nehru National Urban Renewal Mission (JNNURM)

Jawaharlal Nehru National Urban Renewal Mission (JNNURM) is a central scheme for taking up infrastructure projects including garbage, sewerage, parking lots, desilting and beautification of St. Inez nallah etc. in Panjim town. The Community Development Plan of Panjim town has been prepared and sent to Government of India for approval. The Detail Project Reports (DPRs) are being prepared by Corporation of the City of Panaji.

b. Integrated Development of Major Towns (IDMT)

Under the scheme, the State Government provides assistance for accelerating the pace of development of infrastructure in Municipal towns. The scheme is implemented through Goa State Urban Development Agency (GSUDA). The Agency in consultation with Urban Local Bodies, NGOs and professional groups has identified various projects like beautification of areas in Municipal Council jurisdiction, construction of footpaths, development of parks/traffic junctions, procurement of sweeping machines for Panaji, Margao, Mormugao, Curchorem, Cacora and Mapusa Municipal Councils etc.

Achievements

- Various seminars were conducted at State and Town level. Besides NGOs, the staff of Municipal Councils and members of C.D.S participated. Training and awareness programmes were conducted by this agency for the group members of C.D.S of every Municipal area so that they could derive the benefits of the schemes in various components like subsidy for self-employment, wage employment and up gradation of skills etc.
- The Agency has entered into a MoU with Fr. Agnel Ashram to impart the skill up gradation programme in various commercial trades to help the B.P.L. beneficiaries to come out with self entrepreneurship by availing the subsidies provided by this Agency.
- Grants are released to Municipal authorities to meet the expenditure on operation/ maintenance of street lights, construction/maintenance of roads, drains, construction of public toilets etc. The work of ACC Marg on pilot basis is taken up in the jurisdiction of Cuncolim Municipal Council.
- Solid Waste Management Plan is introduced in the Municipal Areas in order to provide clean environment and freedom from menace of garbage. There is a proposal to establish two modern solid disposal plants which will cater to

the needs of Councils and Village Panchayats.

B. FIRE AND EMERGENCY SERVICES

Achievements

- The first batch of BSC Fire Technology of 3 $^{1/2}$ years duration have completed the course. Goa is the only State in India where a degree course in Fire Technology is conducted in collaboration with the State Fire Department. The course is a good example of Private Public participation in developmental activities.
- 1443 candidates have been imparted training in Elementary Fire Prevention and Fire Fighting sponsored by Private/Government agencies from different parts of the country.
- Under Calamity Relief Fund, an amount of Rs.23.98 lakh have been provided during 2007-08 for purchase of Search & Rescue Equipment. Tenders have been floated and the process of acquiring will be completed shortly.
- Revenue of the department is generated by way of Fire cess which is levied towards NOC for high rise buildings etc. A Fire Tax is to be introduced by amending suitably the Fire Force Act, 1986, which is being examined. Enhancement of the fee for training candidates from other States/UTs, Government and Private Industries in fire prevention, protection and fire fighting is under examination
- The renovation work of the fire station building at Old Goa has been completed. Proposal for acquiring land for new premises for the station has also been sent to Government.
- The construction work of Calangute fire station is in progress.
- Construction works of dormitory and residential quarters for the staff of the Directorate of Fire and Emergency Sevices, St.Inez, Panaji is proposed to be carried out shortly.
- Construction work of Modern Fire station building at Kundaim Industrial Estate has been entrusted by GIDC to a Contractor. The department will be providing required manpower and equipment for this station.
- Proposal for land acquisition for a modern Fire Station at Curchorem is under consideration of Government.
- During the year, 1961 fire and emergency calls were attended and 143 human/animals and property worth Rs.17.61 crore was saved/salvaged. 1140 prevention inspections were carried out and Rs. 10.00 lakh was collected towards licence fee.
- During the year, an amount of Rs.14.00 lakh has been remitted to the Government Treasury towards training charges.

Besides providing fire protection cover, fire prevention and fire fighting, the department is now focusing on fire safety underlying the need for proactive measures. Awareness is sought to be generated by imparting various

types of fire fighting and fire prevention training to in service personnel and tailor made courses for the members of the public and security agencies so as to raise awareness in fire safety practices. The Goa Fire Service has been identified as Regional Training Centre by the Ministry of Home Affairs, Government of India for imparting professional career courses on All India basis

C. TOWN & COUNTRY PLANNING DEPARTMENT

Achievements

- A Task Force for Regional Plan constituted for chalking out a road map for preparation of the Regional Plan 2021 is in process of finalizing its preliminary report.
- The Goa (Regulation of Land Development and Building) Ordinance 2007, framed, is under submission and approval of the Government.
- Goa Tax on Infrastructure Ordinance 2007 was re-promulgated for levying higher Infrastructure Tax [Rs. 30/- to Rs. 40/- per M^2] on multi dwelling units and including industrial buildings under the Tax Net Infrastructure development in the State.
- A workshop was organized by the Department on the review of Town & Country Planning Act, 1974 and the process and interactions towards overhauling the Act has been initiated.
- Effective Development Control work as advisory to local bodies and implementation of section 17-A of TCP Act.
- Regional Workshop on the National Urban Information System scheme was organised by the department in association with Town & Country Planning Organization, Government of India and the North Goa PDA, Panaji. The Town & Country Planning Department is the State Nodal Agency and Mormugao, Margao, Mapusa, Cuncolim & Curchorem-Cacora towns are included in the scheme and the Capacity Building Programme of staff at the State Nodal Agency and local bodies and statistical survey works are being taken up.
- The ODPs for 5 planning areas are prepared and their revision is under process.
- Zoning Plans for many towns and fast growing urban areas are prepared and are being implemented.
- 29 Industrial estates have been developed by GIDC as planned.
- Layout Plans of various Industrial areas such as Pilerne Industrial Estate and Phase II of Verna Industrial area etc. are planned and designed by the department and are implemented/ developed by GSIDC.
- Satellite towns for Panaji i.e. Kadamba Plateau, Bambolim/Dona-Paula, Alto-Porvorim planned as outgrowths of Panaji area are now developed as major urban areas which has absorbed the spillover growth of Panaji and eased the Capital City's problems to great extent.

- Various development through PDAs schemes are being taken up in major towns of Panaji, Margao, Vasco and Ponda. Bye-passes proposed for towns of Mapusa, Panaji, Ponda, Vasco etc, are being developed.
- Under IDSMT (Central scheme), residential area at Porvorim, Ponda, Vasco, Corlim and few other places have been developed.
- A water quality study of Mandovi and Zuari river basin was conducted.
- An ecological study of coastal areas and a status report on the development of beaches within 500 mts. from High Tide Line (HTL) which has been the basis for evolving guidelines for control of construction in the ecologically sensitive beach areas was conducted.
- A study of the estuarine area of Mandovi and Zuari river area was conducted and a status report was prepared for the Ecological Development Council of Goa which helped in framing development guidelines for estuarine area development.
- Based on the study carried on Conservation and Preservation of Architectural heritage of Goa, the Conservation Areas were marked in ODP of Panaji/Margao and the "Conservation Committee" was constituted to consider developments in these areas.
- Preparation of Infrastructure development plans for Konkan Railway stations.
- Preparation and implementation of Outline Development Plan for towns of Margao, Panaji, Vasco, Mapusa, Ponda and Taleigao. Preparation and implementation of detail Zoning Plan for Valpoi, Sanquelim, Quepem, Canacona, Curchorem-Cacora, Cuncolim, Sanguem, etc.
- Traffic Circulation Plans and intersection designs for major towns and junctions were taken up and submitted to Traffic Police and PWD for implementation.

The Goa [Regulation of Land Development and Building Construction] Ordinance 2007 was promulgated which enables the Government to notify the uniform regulation throughout Goa with respect to zoning of the land and its use, general requirements for land development, group housing, sub-division of land including amalgamation of plots, preservation and conservation, procedure for seeking development permission and any other aspect relating to planning and development which is a land mark legislation in the State of Goa. This will help in bringing simplified licensing procedure and orderly development of the towns and countryside. Regulations framed is under submission and approval of Government.